PAREENITA A.SHIRSATH B.E.A.I.&.D.S. ROLL.NO:49

SMA EXPERIMENT : 06

```
import re
!pip install google-api-python-client networkx pandas matplotlib
import pandas as pd
import networkx as nx
import matplotlib.pyplot as plt
from googleapiclient.discovery import build
API_KEY = "AIzaSyDFZViTcfxxjFMr67zLyr82Z7ylElnipKg"
VIDEO_URL = "https://youtu.be/5PrZvPeUw60?si=BrgIvCfH1--cFd93"
# Extract the video ID from the URL
match = re.search(r'(?:youtube\.com/watch\?v=|youtu\.be/)([\w-]+)', VIDEO_URL)
if match:
    VIDEO_ID = match.group(1)
else:
    VIDEO_ID = None # Or raise an error, or handle invalid URL

if VIDEO_ID:
    youtube = build("youtube", "v3", developerKey=API_KEY)
    comments = []
    authors = []
    request = youtube.commentThreads().list(
     part="snippet",
     videoId=VIDEO_ID,
     maxResults=100
    )
    response = request.execute()
    for item in response['items']:
     comment = item['snippet']['topLevelComment']['snippet']['textDisplay']
     author = item['snippet']['topLevelComment']['snippet']['authorDisplayName']
     comments.append(comment)
     authors.append(author)
    df = pd.DataFrame({
     "Author": authors,
     "Comment": comments
    })
    print("Sample Data")
    print(df.head())
    G = nx.Graph()
    for user in df["Author"]:
     G.add_node(user)
    for i in range(len(authors)-1):
     G.add_edge(authors[i], authors[i+1])
    degree_centrality = nx.degree_centrality(G)
    betweenness_centrality = nx.betweenness_centrality(G)
    closeness_centrality = nx.closeness_centrality(G)
    centrality_df = pd.DataFrame({
     "User": list(degree_centrality.keys()),
     "Degree Centrality": list(degree_centrality.values()),
     "Betweenness Centrality": list(betweenness_centrality.values()),
     "Closeness Centrality": list(closeness_centrality.values())
    })
    print("\nCentrality Analysis")
    print(centrality_df.sort_values(by="Degree Centrality", ascending=False))
    top_user = max(degree_centrality, key=degree_centrality.get)
    print("\nMost Influen0al User:", top_user)
    plt.figure(figsize=(10,7))
    pos = nx.spring_layout(G)
    nx.draw(
     G,
     pos,
     with_labels=True,
     node_color="lightblue",
     node_size=1500,
     edge_color="gray",
     font_size=8
    )
    plt.title("YouTube Comment Interaction Network")
    plt.show()
else:
    print("Could not extract video ID from the provided URL.")
```

```
Requirement already satisfied: google-api-python-client in /usr/local/lib/python3.12/dist-packages (2.192.0)
Requirement already satisfied: networkx in /usr/local/lib/python3.12/dist-packages (3.6.1)
Requirement already satisfied: pandas in /usr/local/lib/python3.12/dist-packages (2.2.2)
Requirement already satisfied: matplotlib in /usr/local/lib/python3.12/dist-packages (3.10.0)
Requirement already satisfied: httplib2<1.0.0,>=0.19.0 in /usr/local/lib/python3.12/dist-packages (from google-api-python-client) (0.31.2)
Requirement already satisfied: google-auth!=2.24.0,!=2.25.0,<3.0.0,>=1.32.0 in /usr/local/lib/python3.12/dist-packages (from google-api-python-client) (2.47.0)
Requirement already satisfied: google-auth-httplib2<1.0.0,>=0.2.0 in /usr/local/lib/python3.12/dist-packages (from google-api-python-client) (0.3.0)
Requirement already satisfied: google-api-core!=2.0.*,!=2.1.*,!=2.2.*,!=2.3.0,<3.0.0,>=1.31.5 in /usr/local/lib/python3.12/dist-packages (from google-api-pythc
Requirement already satisfied: uritemplate<5,>=3.0.1 in /usr/local/lib/python3.12/dist-packages (from google-api-python-client) (4.2.0)
Requirement already satisfied: numpy>=1.26.0 in /usr/local/lib/python3.12/dist-packages (from pandas) (2.0.2)
Requirement already satisfied: python-dateutil>=2.8.2 in /usr/local/lib/python3.12/dist-packages (from pandas) (2.9.0.post0)
Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.12/dist-packages (from pandas) (2025.2)
Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.12/dist-packages (from pandas) (2025.3)
Requirement already satisfied: contourpy>=1.0.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (1.3.3)
Requirement already satisfied: cycler>=0.10 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (0.12.1)
Requirement already satisfied: fonttools>=4.22.0 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (4.62.1)
Requirement already satisfied: kiwisolver>=1.3.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (1.5.0)
Requirement already satisfied: packaging>=20.0 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (26.0)
Requirement already satisfied: pillow>=8 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (11.3.0)
Requirement already satisfied: pyparsing>=2.3.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (3.3.2)
Requirement already satisfied: googleapis-common-protos<2.0.0,>=1.56.3 in /usr/local/lib/python3.12/dist-packages (from google-api-core!=2.0.*,!=2.1.*,!=2.2.*,
Requirement already satisfied: protobuf<7.0.0,>=4.25.8 in /usr/local/lib/python3.12/dist-packages (from google-api-core!=2.0.*,!=2.1.*,!=2.2.*,!=2.3.0,<3.0.0,>
Requirement already satisfied: proto-plus<2.0.0,>=1.22.3 in /usr/local/lib/python3.12/dist-packages (from google-api-core!=2.0.*,!=2.1.*,!=2.2.*,!=2.3.0,<3.0.0
Requirement already satisfied: requests<3.0.0,>=2.20.0 in /usr/local/lib/python3.12/dist-packages (from google-api-core!=2.0.*,!=2.1.*,!=2.2.*,!=2.3.0,<3.0.0,>
Requirement already satisfied: pyasn1-modules>=0.2.1 in /usr/local/lib/python3.12/dist-packages (from google-auth!=2.24.0,!=2.25.0,<3.0.0,>=1.32.0->google-api-
Requirement already satisfied: rsa<5,>=3.1.4 in /usr/local/lib/python3.12/dist-packages (from google-auth!=2.24.0,!=2.25.0,<3.0.0,>=1.32.0->google-api-python-c
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packages (from python-dateutil>=2.8.2->pandas) (1.17.0)
Requirement already satisfied: pyasn1<0.7.0,>=0.6.1 in /usr/local/lib/python3.12/dist-packages (from pyasn1-modules>=0.2.1->google-auth!=2.24.0,!=2.25.0,<3.0.0
Requirement already satisfied: charset_normalizer<4,>=2 in /usr/local/lib/python3.12/dist-packages (from requests<3.0.0,>=2.20.0->google-api-core!=2.0.*,!=2.1.
Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.12/dist-packages (from requests<3.0.0,>=2.20.0->google-api-core!=2.0.*,!=2.1.*,!=2.2.*,!=
Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python3.12/dist-packages (from requests<3.0.0,>=2.20.0->google-api-core!=2.0.*,!=2.1.*,!=2.
Requirement already satisfied: certifi>=2017.4.17 in /usr/local/lib/python3.12/dist-packages (from requests<3.0.0,>=2.20.0->google-api-core!=2.0.*,!=2.1.*,!=2.
Sample Data
              Author                                        Comment
0     @amlanmohanty1  🚨Are you building your Data Analyst Portfolio?...
1  @SUNILYadav-tl3ug  Very good....I am engineer Mechanical Engineer...
2   @kajoljituri4501  But  you did not show how you extracted the da...
3     @karthik0803-s  I upload your 2 projects in my resume one is z...
4     @VIRAJV.HADKAR  Really enjoyed this video—great explanation ov...

Centrality Analysis
                   User  Degree Centrality  Betweenness Centrality  \
26    @kartikeysinha4685           0.073684                0.446585
1      @SUNILYadav-tl3ug           0.021053                0.021053
3         @karthik0803-s           0.021053                0.061814
2       @kajoljituri4501           0.021053                0.041657
6            @killer7495           0.021053                0.119597
..                   ...                ...                     ...
91     @hr_vibe_official           0.021053                0.081523
94     @vaibhavkathole18           0.021053                0.021053
0         @amlanmohanty1           0.010526                0.000000
27  @cudjoeaugustine3835           0.010526                0.000000
95          @Shabbeer886           0.010526                0.000000

    Closeness Centrality
26              0.037037
1               0.022274
3               0.023267
2               0.022765
6               0.024837
..                   ...
91              0.023086
94              0.021665
0               0.021794
27              0.035728
95              0.021210

[96 rows x 4 columns]

Most Influen0al User: @kartikeysinha4685
```

YouTube Comment Interaction Network